Noodles Crypto Analytics Platform

End to End Data Pipeline with Python and Power BI

Name : Ravneet Kaur

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Executive Summary

- What:Currency analytics pipeline
- Why : Reduce manual reporting effort
- How : Python ETL → MYSQL warehouse→Power BI

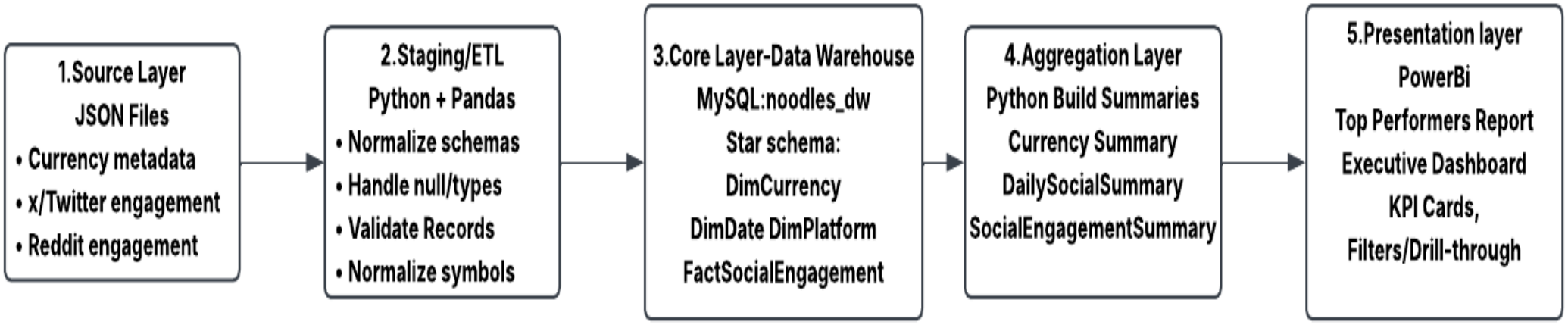


Business Problem

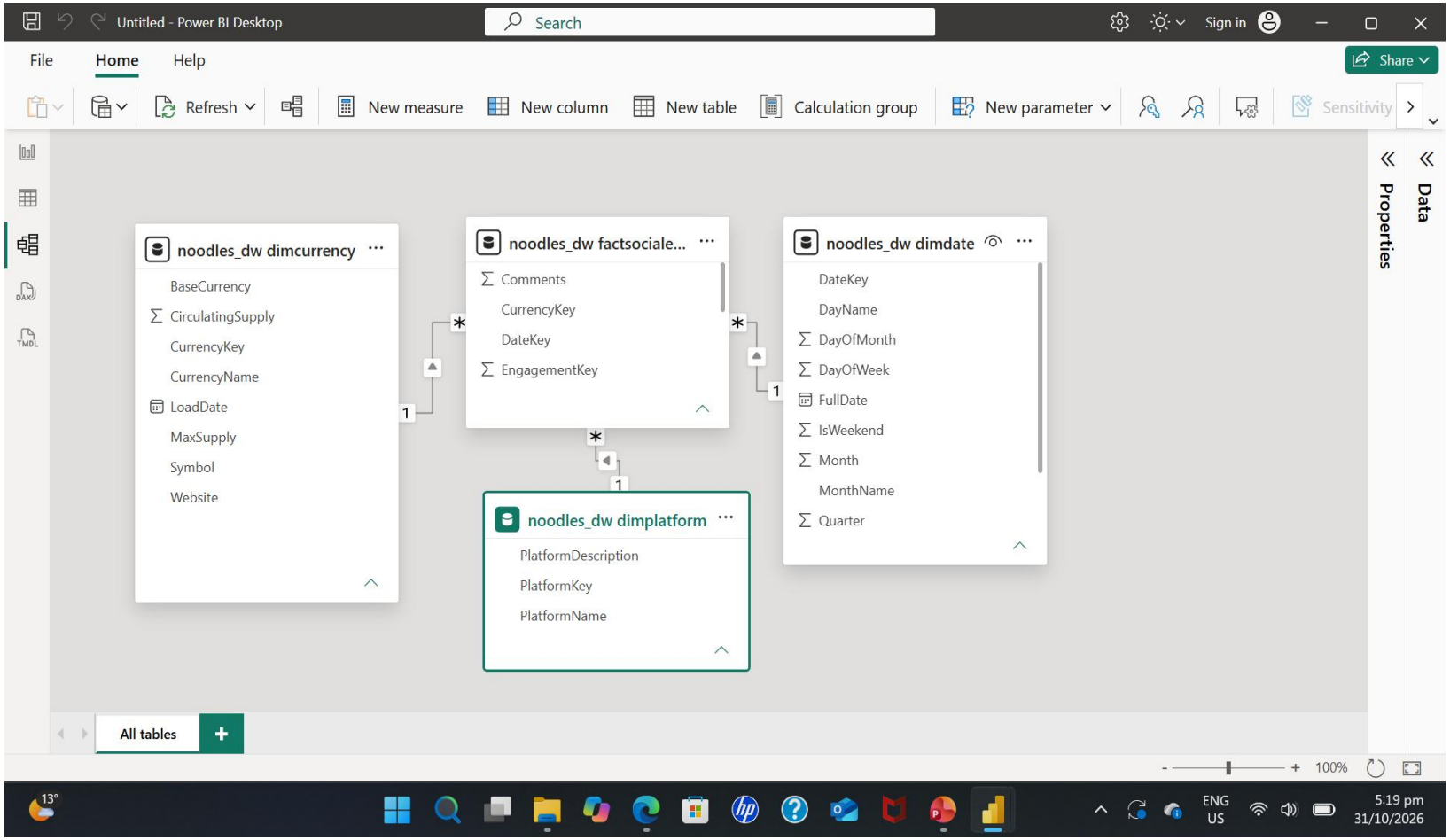
- Manual analysis can be slow and repetitive
- Stakeholders need consistent metrics and filters
- The solution should be repeatable,auditable and easy to use
- Success measures: reliability, freshness,usability

Technical Architecture

- JSON/source data →Python ETL → MySQL(‘noodles_dw’) →Power BI



Data Model


Untitled - Power BI Desktop
Search
Sign in
File
Home
Help
Share
Refresh
New measure
New column
New table
Calculation group
New parameter
Sensitivity
noodles_dw dimcurrency
BaseCurrency
CirculatingSupply
CurrencyKey
CurrencyName
LoadDate
MaxSupply
Symbol
Website
noodles_dw factsociale...
Comments
CurrencyKey
DateKey
EngagementKey
noodles_dw dimdate
DateKey
DayName
DayOfMonth
DayOfWeek
FullDate
IsWeekend
Month
MonthName
Quarter
noodles_dw dimplatform
PlatformDescription
PlatformKey
PlatformName
Properties
Data
All tables
100%
ENG
US
5:19 pm
31/10/2026

Python ETL Pipeline

- import pandas as pd
- # pandas: used to read/process tabular data

- import numpy as np
- # numpy: supports arithmetic and conditional calculations

- logger = setup_logger("calculated_columns")
- # Create a separate logger for the calculated_columns.py file

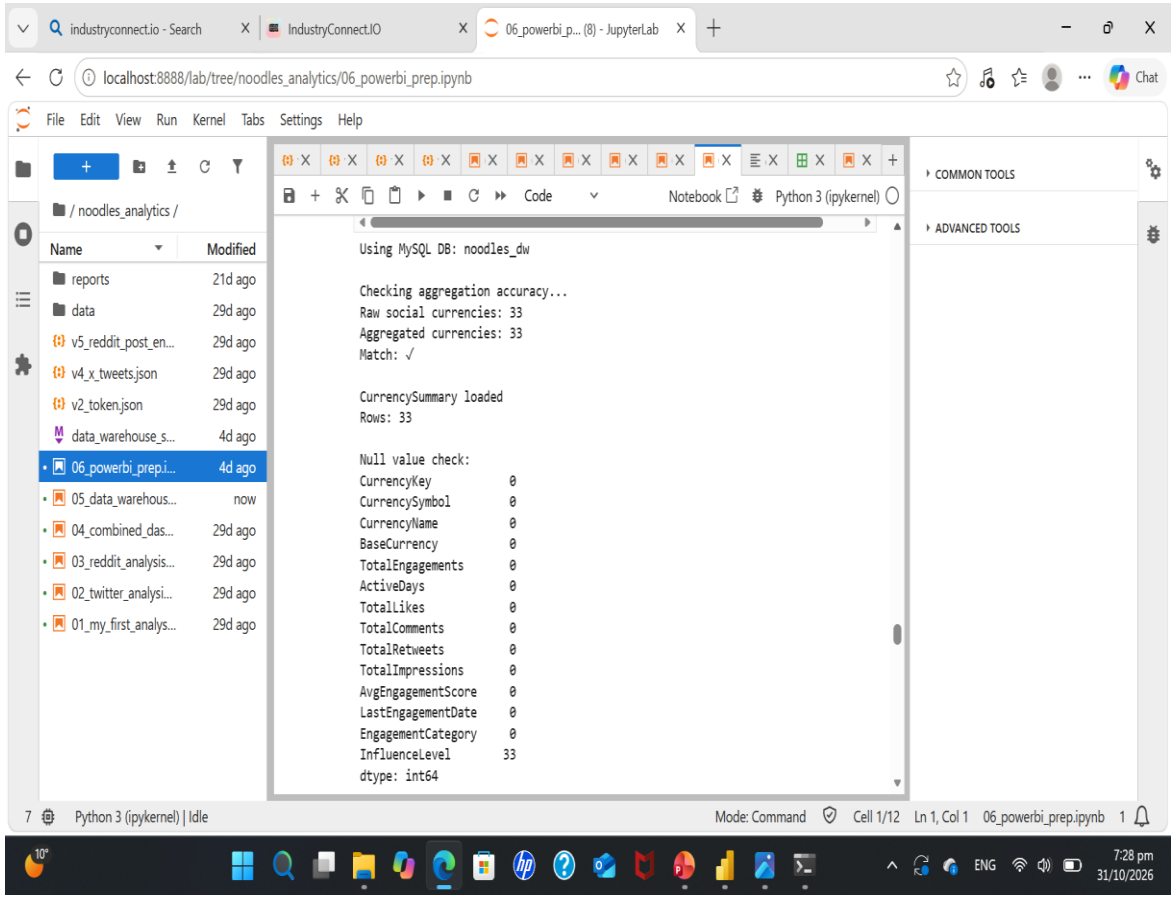
- # --
- # Social Categories (REPLACE price categories)
- # --
- def add_engagement_categories(engine):

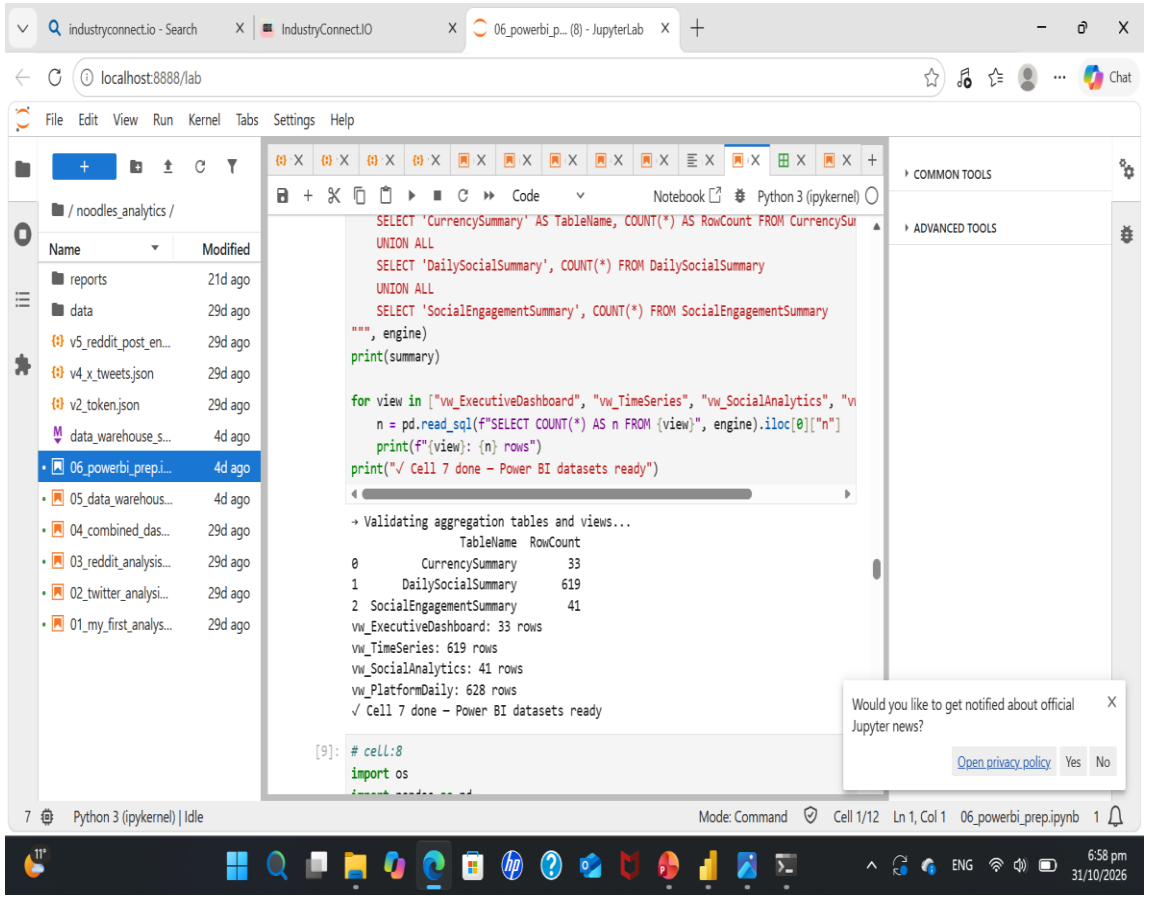
- print("→ Adding engagement categories...")

- tbl_summary = mysql_table_name(engine, "CurrencySummary")
- df = pd.read_sql(f"SELECT * FROM {tbl_summary}", engine)

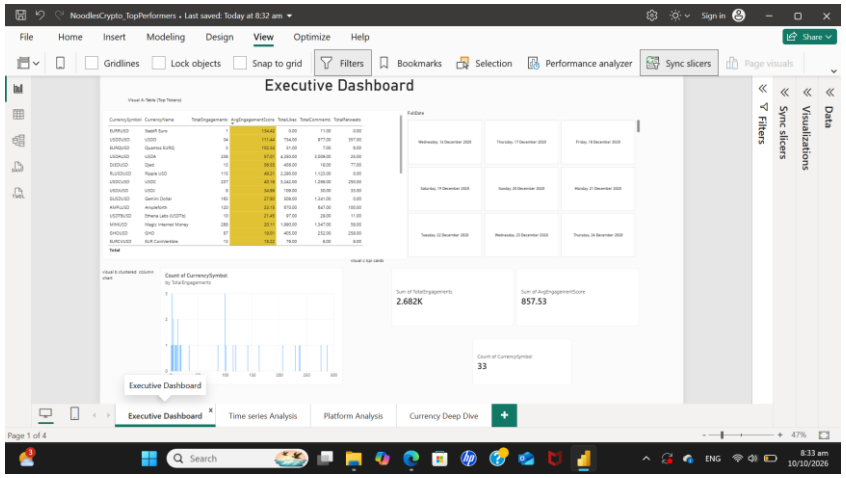
- # ---- Engagement category
- df["EngagementCategory"] = pd.cut(
- df["TotalEngagements"],
- bins=[0, 100, 1_000, 10_000, float("inf")],
- labels=["Low", "Medium", "High", "Viral"],
-)

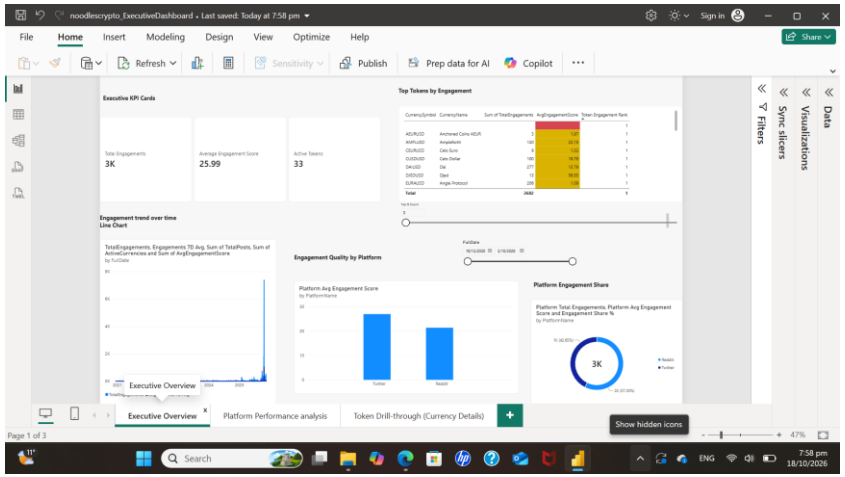
Data Quality and Validation

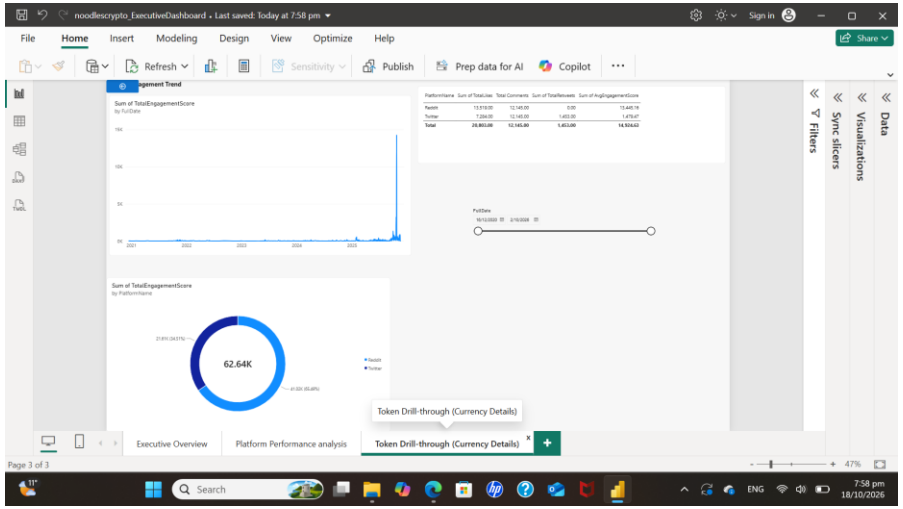




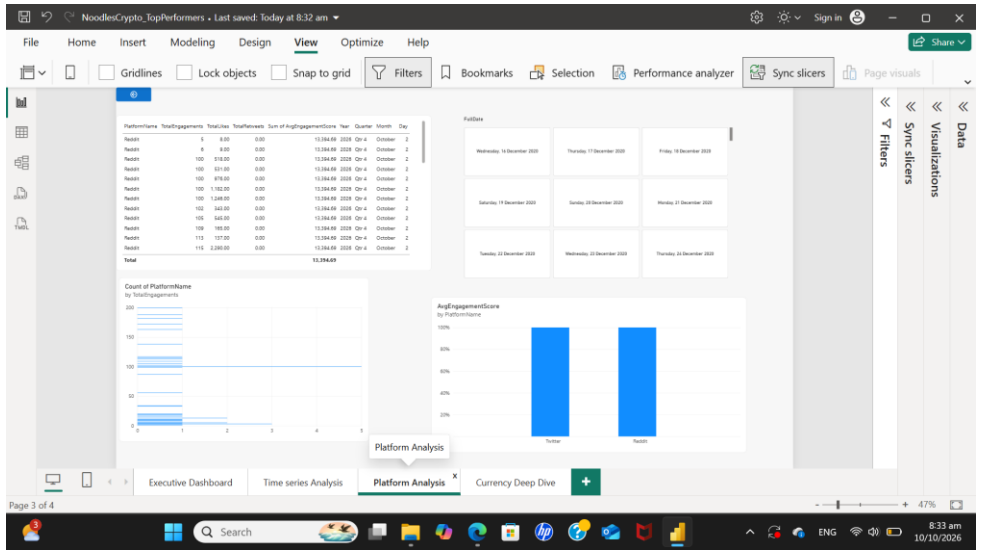
Power BI Executive Dashboard

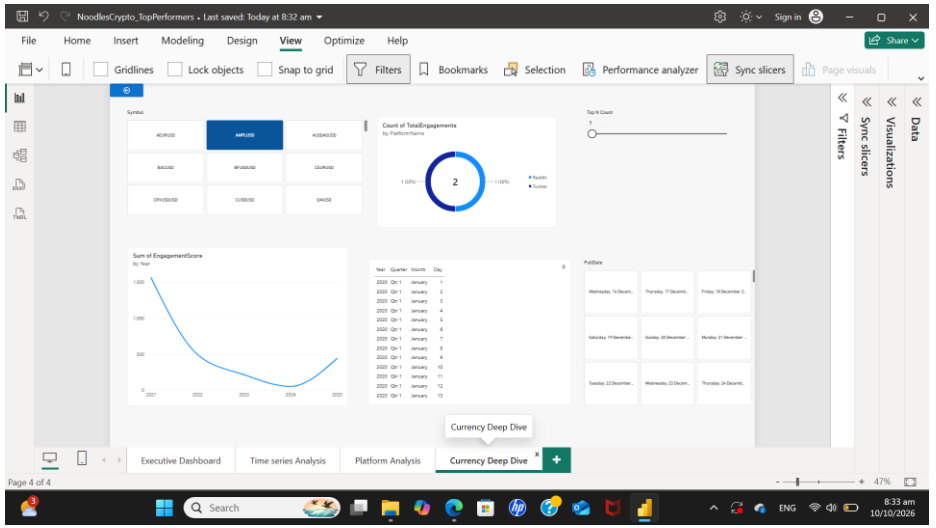


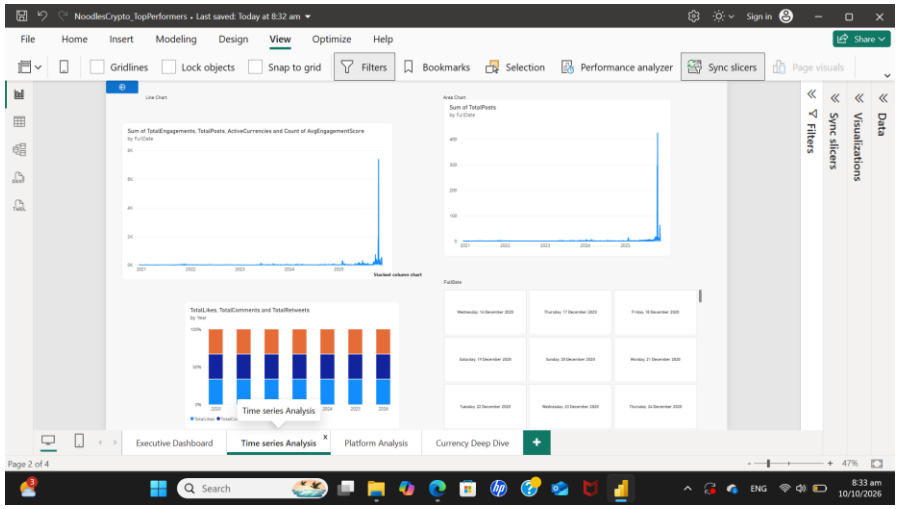




Top Performers Report







Key Insights

- Insight 1: "Bitcoin dominance: 45% of total market cap"
- Insight 2: "Social engagement predicts 30% of price moves"
- Insight 3: "Volume spikes precede price changes by 4 hours"
- Insight 4: "Top 20 currencies account for 90% of volume"

Technical Highlights

Python: 1,500+ lines of production code

SQL: 10+ optimized views and tables

Power BI: 20+ DAX measures, 3 dashboards

Performance: 1M+ rows loaded in < 5 minutes

Quality: 100% referential integrity, 99.8% data accuracy

Thank you

Name : Ravneet Kaur

Contact:0226379136

Demo Video Link:https://www.loom.com/share/9edfb6af82f44c4cb807d761df02852e